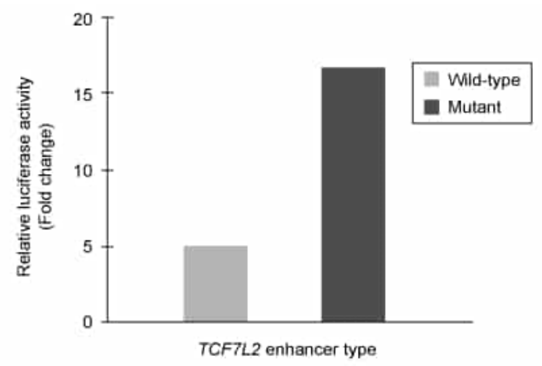
A cytosine-to-thymine mutation in the regulatory region of the *TCF7L2* gene is associated with an increased risk of developing type 2 diabetes mellitus (T2DM). The figure below shows an expression profile using a luciferase-based reporter gene attached to wild-type and mutated regulatory enhancers for *TCF7L2*.

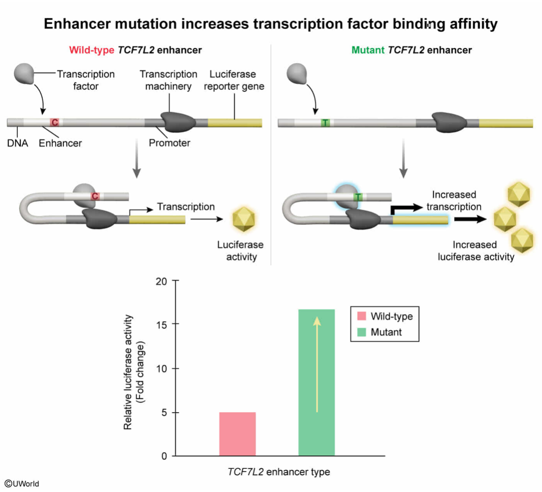


Which of the following most likely explains how this mutation could lead to an increased risk for T2DM?

- A. The binding of a transcriptional activator is increased when the mutation is present. [71%]
- B. The activity of a transcriptional repressor is increased by the mutation. [8%]
- C. The *TCF7L2* enhancer becomes methylated, increasing activation of transcription. [12%]
- D. The acetylation of histones around the *TCF7L2* gene is decreased, making it transcriptionally inactive. [7%]

Correct
71% answered correctly

Explanation:



Enhancers are DNA regulatory elements located near genes that function in the transcriptional control of gene expression in eukaryotes. Enhancers are found mostly in noncoding regions, between two genes or within introns, sometimes at great distances from the target gene. Enhancer sequences contain clusters of specific transcription factor binding sites that are bound by cell-type specific transcription factors, chromatin modifiers, co-regulators, and RNA polymerase. When regulatory proteins assemble at the enhancer, a loop is formed, which allows for physical contact with the promoter of the target gene. These interactions activate the initiation of transcription by engaging RNA polymerase.

Transcription activators are proteins that bind to enhancers and contact the transcriptional machinery to promote transcription. Transcriptional repressors can also bind to enhancers to block activation. Transcriptional activators and repressors can also affect chromatin organization, recruiting proteins that acetylate or deacetylate histones promoting or repressing transcription of the gene.

The given graph depicts an increase in transcription of a luciferase reporter gene when the *TCF7L2* enhancer mutation is present, which is consistent with an **increase in binding of a transcriptional activator**. When the mutation is **present**, transcriptional activity is significantly **greater** than normal. Therefore, increased expression of the *TCF7L2* gene is likely associated with the increased risk for type 2 diabetes.

(Choice B) The given graph demonstrates that expression of the reporter gene increases when the mutation is present; however, increased binding of a transcriptional repressor would cause a decrease (*not* increase) in *TCF7L2* expression.

(Choice C) DNA methylation is associated with decreased (*not* increased) transcriptional activity. In addition, when DNA is methylated cytosine is the nucleotide that is typically affected rather than thymine. In this case, the mutation replaces cytosine with thymine, so increased DNA methylation would *not* occur at that position.

(Choice D) Acetylation of histones results in increased transcriptional activity. Decreasing the number of acetyl groups on the histones would decrease (*not* increase) transcription.

Educational objective:
Gene regulation in eukaryotes is mediated by DNA regulatory sequences called enhancers, located outside the coding sequence. Transcription can be activated or repressed by a transcription factor binding to enhancers, which allows for the differential expression of genes. Enhancers also modulate transcriptional activity by influencing chromatin organization.

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